

GLASGOW COLLEGE UESTC

Main Exam

Engineering Project Management & Finance (UESTC 3031& UESTCHN3012)

**Date: Sunday (June.22nd, 2025)
Time: 14:30-16:30**

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

For Multiple Choice Questions, use the dedicated answer sheet provided.

**Make sure that your University of Glasgow and UESTC Student Identification
Numbers are on all answer sheets.**

**An electronic calculator may be used provided that it does not allow text storage
or display, or graphical display.**

**All graphs should be clearly labelled and sufficiently large so that all elements
are easy to read.**

**The numbers in square brackets in the right-hand margin indicate the marks
allotted to the part of the question against which the mark is shown. These
marks are for guidance only.**

Section A: Multiple Choice Questions

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options for each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

- (1.1) Which item below is correct regarding Design for Manufacturing (DFM): [2]
- A. As a general rule, DFM can't be started at the initial stages of product design before fabrication.
 - B. DFM has advantages, including lower production costs and higher product quality; however, it may increase the time required to bring the product to market.
 - C. The DFM may impose higher manufacturing costs, but instead, it can benefit from higher customer satisfaction because of improved product quality.
 - D. DFM is a design method to reduce the complexity of manufacturing operations and the overall cost of production including the cost of raw materials.
 - E. While manufacturers were facing strong competition from other manufacturers (rivals), at the time, they were under pressure to increase production efficiency while increasing costs of production.
- (1.2) There is a variety of actionable sustainable design principles that can be implemented to create better designs. Some of them are below except: [2]
- A. Dematerialization.
 - B. Migration to product-service systems.
 - C. Limit or eliminate long-distance outsourcing.
 - D. Design for a shorter period of usage.
 - E. Invest in simulation.
- (1.3) Which of the following 7QC tools can best be used to logically organize possible factors related to a specific problem by graphically displaying them in increasing detail and suggesting causal relationships among theories? [2]
- A. Cause-Effect Diagram.
 - B. Check Sheet.
 - C. Pareto Chart.
 - D. Control Charts.

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E. Scatter Diagram.

- (1.4) Which model below is best used as a roadmap for Six Sigma to improve the quality of results produced by company processes? [2]
- A. Define, Measure, Analyze, Improve and Control (DMAIC)
 - B. Plan, Do, Check, Act (PDCA)
 - C. Total Quality Management (TQM)
 - D. Lean Manufacturing
 - E. Failure Modes and Effects Analysis (FMEA)
- (1.5) Project A has an NPV of \$1.2 million; Project B has a BCR of 1.035; Project C has an IRR of -0.03 percent. Based on the information given, which project appears to be most attractive? [2]
- A. Project A is the most attractive.
 - B. Project B is the most attractive.
 - C. Project C is the most attractive.
 - D. All three projects A, B, and C are equally attractive to execute.
 - E. It is uncertain which project is the most attractive to execute based on the provided information.
- (1.6) You are a project manager for an electrical manufacturing company involved in developing innovative components for renewable energy solutions. You have noticed that one of your team members is lacking knowledge in electronic component design. The same was communicated to him, and you tried sending him for training. However, the team member started arguing that he is an expert in electronic component design. Which of the following best describes this situation? [2]
- A. The team member is consciously competent.
 - B. The team member is unconsciously competent.
 - C. The team member is consciously incompetent.
 - D. The team member is unconsciously incompetent.
 - E. The team member is aware of his/her skill, so there is no necessity for training.
- (1.7) In a Profit and Loss (Income) statement, Earnings before Interest, Tax, Depreciation, and Amortization (EBITDA) is equal to: [2]

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A – Total Sales minus Cost of Sales

B – Total Expenses minus Net Profit

C – Total Sales minus (Salaries + Rent)

D – Gross Profit minus all company operating expenses (salaries and rent etc.)

E – Full production costs minus Net Profit

- (1.8) You are an engineer working on a project that depends on a new type of battery technology for success. During the development of the product, a report appears in a respected science publication that there may be health issues associated with the new battery technology if it leaks but they do not yet have reliable data. You are under pressure from your boss to complete your project, what do you do? [2]

A – Tell your boss there is no reliable data so you should complete the project against the original plan. It is not your fault if there is no reliable data.

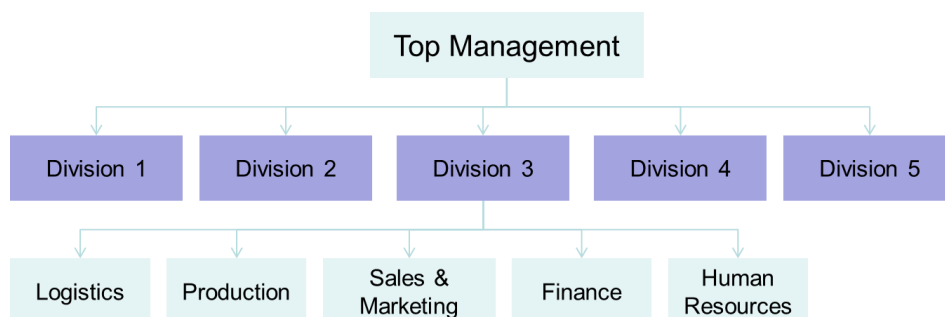
B – Advise your boss that although there is no reliable data, you should adopt a precautionary approach and install further protection (costing more money) around the battery along with a monitoring system to detect leaks.

C – Stop work on the project and advise your boss that if he does not cancel the project you will report him to the health authorities.

D – Advise your boss if you cannot use the battery technology then they should cancel the project and the product that uses it as it will not be competitive in the market.

E – Approach the author of the science publication and offer to pay for a holiday for them and their family if the author writes another article stating that the first article was wrong.

- (1.9) If a company has an organizational structure shown below, it is called: [2]



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- A – A Team-Based Structure
- B – A Matrix Structure
- C – A Divisional Structure
- D – A Simple Startup Structure
- E – A Military Command Structure

(1.10) You have fifteen members in your engineering project team. You have called for a project status meeting without prior notice. During the meeting, there were too many side conversations and arguments. As a result, the team achieved very little in the meeting. To achieve orderliness in the meeting next time, you should: [2]

- A – Decrease the number of people in the meeting
- B – Publish a meeting agenda
- C – Ensure that you control the channels of communication
- D – Give incentives to team members for conforming to desired meeting norms
- E – Invite a member from the higher management/executive team

(1.11) A part of the cash flow with 30-days net payment terms for cash-in and cash-out is given in Table Q1 (1.11) below. Determine the cumulative cash towards the end of each month (M). The currency utilized for transactions in this cash flow is RMB. [2]

Table Q1 (1.11): Cash-Flow Table

	M0	M1	M2	M3	M4	M5
Sales Bookings	10	12	8	20	100	0
Shipments	0	10	12	8	20	100
Components Order	5	6	4	10	50	0

A – M0 = 0 RMB; M1 = -5 RMB; M2 = -1 RMB; M3 = +7 RMB; M4 = +5 RMB; M5 = -25 RMB

B – M0 = -5 RMB; M1 = -6 RMB; M2 = -4 RMB; M3 = -10 RMB; M4 = -50 RMB; M5 = 0 RMB

C – M0 = 0 RMB; M1 = -10 RMB; M2 = -12 RMB; M3 = -8 RMB; M4 = -20 RMB; M5 = -100 RMB

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D – $M_0 = -5$ RMB; $M_1 = -16$ RMB; $M_2 = -16$ RMB; $M_3 = -18$ RMB; $M_4 = -70$ RMB; $M_5 = -100$ RMB

E – $M_0 = 0$ RMB; $M_1 = -5$ RMB; $M_2 = +4$ RMB; $M_3 = +8$ RMB; $M_4 = -2$ RMB; $M_5 = -30$ RMB

(1.12) Based on Q1 (1.11) above, how do you classify the trading nature for such a type of cash-flow statements? [2]

A – Well-balanced trading

B – Under trading

C – Over trading

D – No trading

E – Well-established trading

Section B: Long Questions

Q2 [25 marks]

- (a) A company is designing a new household appliance and is considering two possible designs for one of its components.

Design A consists of five separate parts that need to be assembled.

Design B integrates these parts into a single molded component.

- (i) Discuss the advantages and disadvantages of each design from a Design for Manufacturing (DFM) perspective. [6]
 - (ii) Design A consists of five separate parts that need to be assembled, and one of these parts can be reused in other devices. Design B integrates all parts into a single molded component, making it smaller in size. In terms of actionable sustainable design principles, compare the two designs. [4]
 - (iii) The 4Rs, in particular, are a very effective tool in building a circular economy. They include four principles that can be applied by almost every individual and have an exponential effect. Distinguish the principles of the 4Rs that Design A and Design B primarily apply to promote the circular economy. [3]
- (b) The night shift operator recorded twenty current (mA) samples during the electronic circuit testing over time in electronics manufacturing, as shown in Table Q2(b). As a technician, you are expected to evaluate and analyze the process over the last 20 time units to determine whether the process is under control (stable) or not. Discuss your findings by utilizing a control chart (considering $2 * \text{Sigma}$). You are required to create a plot to demonstrate your explanation with a visual representation that includes the central line, upper control limit (UCL), and lower control limit (LCL). [7]

#Sample	Current (mA)	#Sample	Current (mA)
1	510	11	505
2	516	12	511
3	500	13	500
4	495	14	498
5	505	15	501
6	500	16	500
7	510	17	490
8	505	18	505
9	500	19	507
10	499	20	500

Table Q2(b)

- (c) Assume that the manufacturer's target for the process in (b) is 500 mA, with a tolerance of ± 5 mA. To assess the manufacturing process's capability to meet specifications, evaluate both process capability indices for this manufacturing operation. Discuss your findings. [5]

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Q3 [25 marks]

(a) Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements. Answer the following questions.

- (i) John is overseeing the development of a new electronic device prototype. After collaborating with the team, they determine the most likely project duration of 10 months. In the best-case scenario, efficient collaboration with a specialized component supplier could reduce the duration by 2 months. However, a delay in obtaining critical components might extend the project by 3 months. Using the three-point estimating technique, calculate the expected project duration using both Triangular Distribution and Beta Distribution. Also, determine the standard deviation for each of these approaches. [3]
- (ii) You are leading a project team consisting of 11 members. As the project transitions into the next phase, three new members join the team while two existing members depart. Based on these changes, determine the number of additional communication channels that will be created. [2]
- (iii) Assume an electronic product engineer has collected 10 observations (test results) for a specific product's quality characteristic, which is nominally the best at a target value of 8.5. The engineer uses the Taguchi loss function to evaluate the quality, where the constant $k=1$. Given the data provided in Table Q3(a), compute the Taguchi loss function for each observation and determine the total loss. [3]

Observation No.	Measurement (Quality Characteristic)	Observation No.	Measurement (Quality Characteristic)
1	8.1	6	8.35
2	8.9	7	8.25
3	8.45	8	8.68
4	9.25	9	8.9
5	8.86	10	9.05

Table Q3(a)

- (iv) You have been assigned as the project manager for the development of a basic embedded system for controlling a temperature sensor in a smart home application. The major phases of the project are planning, design, and testing. The WBS is decomposed down through these phases, where each phase is broken down into two additional activities at Level 2. Based on this, plot the WBS for the project, showing all three levels clearly. [4]
- (b) Project schedule management includes the processes required to manage the timely completion of the project. One of the common techniques used is CPM (critical path method). The critical path is the sequence of activities that represents the longest path

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through a project, which determines the minimum possible project duration. Consider a project (assume date 0 for project starting) with the network diagram shown in Figure Q3(b) and answer the following questions.

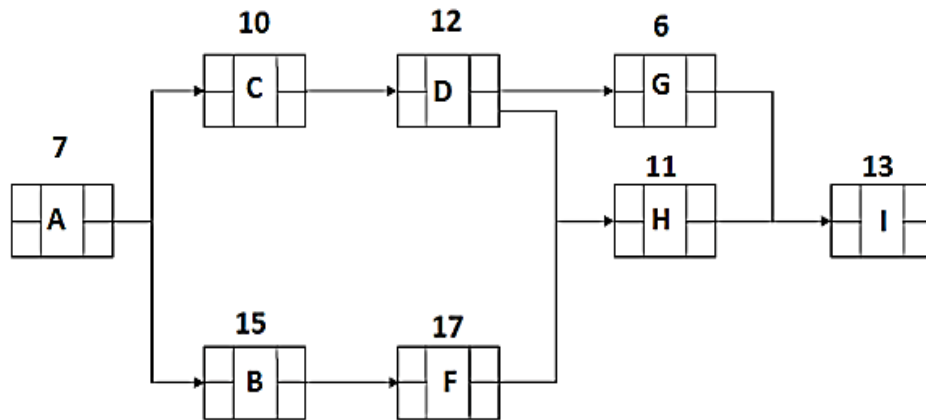


Figure Q3(b)

- (i) Perform the forward pass analysis through the schedule network and compute Early Start and Early Finish dates for all activities. [5]
- (ii) Perform the backward pass analysis through the schedule network and compute Late Start and Late Finish dates for all activities. [5]
- (iii) Calculate the total float (slack) for all activities. [2]
- (iv) Define the Critical Path (sequence of activities that are located on the critical path). [1]

Q4 [26 marks]

VRtuoso is a company that produces virtual reality (VR) headsets for businesses and academic (teaching and research) purposes wherein each headset comprises of two specialized lenses. The company manufactures 0.6 Million (M) headsets annually. The lenses for these headsets are provided by an external supplier to VRtuoso for 675 RMB each. Moreover, the company has some vacant production facilities that adds to the total annual costs with a Fixed Overhead Cost of 39 M RMB. Currently, the vacant production facilities are rented out for 2 M RMB per year but the company has recently been informed during a TV interview that they might not renew the rental agreement for the next term.

- (a) Draw a table comparing the equivalent cost of purchasing a lens from the external supplier while the vacant production facilities are rented out vs. the scenario if the rental agreement for the vacant production facilities is not renewed for the next term. [5]
- (b) VRtuoso drops the plan of renewing the rental agreement for the vacant production facilities instead plans to utilize the vacant production facilities for manufacturing the lenses within the company. In addition to the Fixed Overhead Cost, the production cost breakdown for a lens consists of a direct materials cost of 325.00 RMB, a direct labor cost of 215.00 RMB, and a variable overhead cost of 127.50 RMB. Given this new information, prepare a new cost table demonstrating the annual impact of manufacturing the lenses within the production facilities available in the company. Should VRtuoso continue purchasing the lenses from the external supplier or start manufacturing their own lenses? [5]
- (c) If VRtuoso can sell the VR headsets for 1,413 RMB each, based on the information given above in Q4(b), draw a table demonstrating the production costs and sales revenue of manufacturing and selling VR headsets from 0 → 1 M in quantity with an increment size of 0.2 M VR headsets. [6]
- (d) If the VRtuoso factory is running at full capacity of 0.6 M VR headsets per year with an assumed price of 1,413 RMB per VR headset, what is the profit made by VRtuoso? What is the safety margin for this scenario? [5]
- (e) Draw a Break-even graph showing Total Cost (TC), Variable cost (VC), and Revenue (REV) against production quantity (Q). Calculate the Break-even Point and indicate this on the graph. [5]

End of question paper